A Project Report On

Air Quality Monitoring System Using Iot

Submitted in partial fulfilment for the
degree of Bachelor of Technology in
Artificial Intelligence

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CERTIFICATE

This is to certify that **Arya Gawade, Gungun Gupta, Kajal Naik** has completed the **Phase-II** report on the **"Air Quality Monitoring System Using Iot"** satisfactorily in partial fulfillment for the Bachelor's Degree in **Artificial Intelligence** under the guidance of **Prof.Snehal Bindu** during the year **2025** as prescribed by **Shreemati Nathibai Damodar Thackersey Women's University (SNDTWU)**

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Examiner 2

Abstract

This project focuses on designing and implementing an **IoT-Based Air Quality Monitoring System** that continuously measures and analyzes the surrounding air quality in real time. With the rapid increase in industrialization and urbanization, monitoring air pollution has become a significant environmental issue. Traditional air quality monitoring systems are costly, bulky, and limited to specific locations. To address these issues, this project combines the **Internet of Things (IoT)** with low-cost sensors and microcontrollers to create an affordable, portable, and effective monitoring solution.

The proposed system uses an **ESP8266 NodeMCU** microcontroller as the main processing unit. It connects to a **DHT11** sensor for measuring temperature and humidity, as well as an **MQ135** gas sensor for detecting harmful gases like carbon dioxide (CO), ammonia (NH), and benzene. The collected data is sent via Wi-Fi to the **Arduino IoT Cloud**, where it appears on a real-time online dashboard.

Unlike traditional systems that only show qualitative categories such as Good, Moderate, or Poor, this project provides **detailed numerical readings** and a clear textual message that includes temperature, humidity, and air quality values. For instance: “Temperature = 29°C, Humidity = 62

The project demonstrates how IoT can effectively create a **low-cost, user-friendly, and scalable system** for monitoring the environment. The results indicate that the system tracks changes in environmental conditions accurately, making it suitable for both educational and practical purposes, such as home automation, urban monitoring, and awareness programs.

Keywords: *Air Quality Monitoring, DHT11 Sensor, MQ135 Gas Sensor, Arduino IoT Cloud, Environmental Monitoring*

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Nomenclature

IoT	Internet of Things
ESP8266	Wi-Fi Enabled Microcontroller Used for Data Transmission
NodeMCU	Development Board Based on ESP8266
DHT11	Digital Temperature and Humidity Sensor
MQ135	Gas Sensor Used for Measuring Air Quality
ADC	Analog-to-Digital Converter
AQI	Air Quality Index
PPM	Parts Per Million (Gas Concentration)
PM2.5	Particulate Matter of Diameter 2.5 Micrometers
Wi-Fi	Wireless Network Used for Cloud Connectivity
Cloud	Online Platform Used for Storing and Displaying Sensor Data
Dashboard	IoT Interface for Real-Time Monitoring
MQTT	Message Queuing Telemetry Transport Protocol
Humidity	Amount of Water Vapour in Air (in %)
Temperature	Degree of Hotness of Air (in °C)

Chapter 1

Introduction

Problem statement

Air pollution is increasing rapidly in many urban and industrial areas. Traditional systems for monitoring are very expensive and limited in number, thus unable to provide full real-time data. Consequently, there is a critical lack of timely and effective information about environmental conditions for people and authorities. The demand exists for an affordable, easily deployable, real-time air quality monitoring system able to measure the main environmental parameters and present the data immediately.

Introduction

Cities are growing fast. More cars on the road, more factories running every day—it all adds up to one big problem: air pollution. You see it especially in urban and industrial areas, where the air gets thick with harmful gases and tiny particles. The air quality keeps slipping, and it's not just a distant worry—it's happening now.
The usual way of tracking air pollution relies on big, expensive monitoring stations. These setups cost a lot to install and keep running, so you only find a few scattered around. That means we don't get the full picture. There are gaps in the data, and sometimes, we don't find out about dangerous pollution levels until it's too late. To really protect people and the environment, we need a new approach—something affordable, easy to set up anywhere, and able to keep an eye on the air in real time.
That's where the Internet of Things comes in. Lately, IoT has become a game-changer for environmental monitoring. With connected sensors, cloud storage, and remote access, IoT lets us build air quality systems that don't break the bank. Here's how it

works: sensors like the MQ135 detect harmful gases, the DHT11 keeps track of temperature and humidity, and the ESP8266 handles data collection and transmission. The whole system sends live updates to the Arduino IoT Cloud, so you can see air quality readings instantly, right on a dashboard. This constant stream of data helps people spot trends, react to pollution spikes, and make smarter decisions to stay healthy.

Plenty of researchers have already explored similar setups. They've tried different controllers—Raspberry Pi, ESP32, Arduino—and mixed in sensors like MQ2, MQ7, MQ135, and PM detectors. For sharing and visualizing the data, some used platforms like ThingSpeak, AWS, or Google Cloud. A few even added machine learning to predict air quality index values. Of course, no system is perfect. Issues pop up: sensors can drift over time, everything depends on a stable internet connection, outdoor testing is often limited, and some setups don't measure PM2.5 or PM10. Calibration can be a headache. Still, IoT-based monitoring stands out as a real, low-cost way to spot pollution as it happens.

This project tackles some of those common problems. The system is simple to install, budget-friendly, and works pretty much anywhere—homes, offices, factories, or public spaces. It reacts fast to environmental changes. For example, when exposed to heat, the temperature reading jumps and air quality values drop right away. All the data is available anytime on a cloud dashboard, making it easy to understand what's going on with the air around you.

So, why does this matter? This introduction lays out the need for better air monitoring, explains how the system works, and shows why IoT is such a powerful tool for solving real-world problems.

The last part of this introduction sets up what's next. The following sections dive into the study's goals and break down how the rest of the report is organized.

1.1 Objectives of the Study

1. Build and develop an affordable IoT-based air quality monitoring system.
2. Measure temperature, humidity, and air quality using reliable sensors.
3. Provide real-time monitoring with cloud-based dashboards.
4. Send alerts or notifications when dangerous air pollution levels are detected.
5. Give policymakers, industries, and everyday users useful environmental data they can act on.

Chapter 2

Review of Literature

This chapter presents a detailed review of existing research and IoT-based air quality monitoring systems developed to address the shortcomings of traditional monitoring stations. Conventional air quality stations are accurate but costly, stationary, and limited in coverage, making them unsuitable for continuous, wide-area monitoring. To overcome these limitations, various researchers have proposed IoT-enabled solutions using low-cost sensors, microcontrollers, and cloud platforms to provide real-time environmental data. These systems typically integrate gas sensors, temperature–humidity sensors, wireless communication technologies, and cloud dashboards for analysis and visualization.

2.1 Related Work

[1] **Narute et al. (2025)** developed a real-time air quality monitoring system using a Raspberry Pi, MCP3208 ADC, and MQ-series gas sensors. Their approach involved collecting sensor data every few seconds, preprocessing it, and uploading it to the ThingSpeak cloud. Additionally, a Random Forest model was implemented to predict Air Quality Index (AQI), achieving an accuracy of 85–90%.

[2] **Thirupathamma (2025)** proposed an advanced hybrid system combining stationary and mobile sensing units connected through LoRaWAN, 5G, and Wi-Fi networks. Multiple gas and particulate sensors (CO, NO_2, PM2.5, PM10, SO_2, O_3) were used, while ANN and GRU models were applied for data calibration and forecasting. Another study from IJFMR (2025) emphasized the

lack of coverage and poor predictive capability of conventional monitoring systems, highlighting the need for IoT-based AQI prediction models.

[3] **Agiru et al. (2021)** designed a low-cost IoT solution using MQ2, MQ5, MQ6, and MQ135 sensors with an ESP32 microcontroller. Data was transmitted via MQTT to the Blynk cloud, and a buzzer along with an LCD display provided immediate alerts for unsafe gas levels.

[4] **Munera (2021)** conducted a systematic mapping review of IoT-based air pollution monitoring systems, revealing that most implementations used Arduino or Raspberry Pi, Wi-Fi or LoRaWAN for communication, and cloud platforms for real-time visualization.

[5] **Shankar et al. (2024)** developed a modular system for indoor and outdoor monitoring using MQ sensors and GSM technology, enabling cloud and local storage.

[6] **Jayakumar et al. (2021)** presented a simple IoT solution with Arduino, MQ sensors, and LCD display integrated with a web interface for data monitoring. Together, these studies demonstrate that IoT-based technologies are widely adopted for continuous environmental monitoring due to their scalability, cost efficiency, and real-time data accessibility.

Despite the progress made in the above research, several limitations restrict the performance and scalability of existing IoT-based systems. A common issue across multiple studies is sensor drift, which affects the accuracy of MQ-series gas sensors and requires frequent recalibration. Many solutions depend heavily on Wi-Fi or cloud services, making them unreliable in regions with limited network connectivity, as observed in the works of Narute et al. (2025), Agiru et al. (2021), and Jayakumar et al. (2021). The majority of studies lack particulate matter sensors (PM2.5 and PM10), which are essential for evaluating environmental and health impacts of polluted air. High power consumption, especially with multi-sensor and multi-network systems, restricts long-term outdoor deployment. Machine-learning-based systems proposed in IJFMR (2025) and IJRTI (2025) also require continuous model retraining due to the dynamic nature of environmental conditions. Furthermore, several studies were conducted only on small-scale setups and prototypes, limiting their practical integration into smart city infrastructures. These limitations indicate that existing systems need improvements in pollutant coverage, network flexibility, power optimization, system scalability, and long-term accuracy.

Paper	**Sensors Used**	**Technique / Platform**	**Strengths**	**Limitations / Gaps**
IJFMR (2025)	MQ2, MQ4, MQ7, MQ9, MQ135	Raspberry Pi + MCP3208 + RF + ThingSpeak	This system provides high accuracy ranging between 92–95% and offers very fast alert generation within approximately three seconds.	The system requires regular calibration, depends heavily on network availability, and is prone to sensor drift over time.
IJRTI (2025)	Multi-sensor setup (CO, NO_2, PM2.5, etc.)	LoRaWAN/5G + AI/ML	The model uses AI-based calibration, supports mobile sensing, and can be scaled effectively for larger deployments.	The major limitations include high infrastructure cost and the fact that the model is still in the research and experimental stage.
IJCRT (2021)	MQ6, MQ135	ESP32 + MQTT + Blynk	This system offers real-time alerts and enhances worker safety through continuous monitoring.	Its drawback is the limited industrial coverage which restricts broader applicability.
IJECE (2021)	PM2.5/PM10, CO, NOx, CO2, O3	Arduino + Raspberry Pi + ESP8266 + Cloud	The system is low-cost and capable of performing real-time environmental monitoring.	The accuracy of the sensors is relatively low, and there are only a few large-scale deployments documented.
IJSRA (2024)	MQ2, CO2	Arduino + ESP32 + GSM	The system is suitable for both indoor and outdoor air-quality monitoring.	Its limitations include sensor drift and comparatively high power consumption.
IRJET (2021)	MQ2, MQ7	Arduino + LCD + Webpage	The system is simple to set up and remains low in cost while providing basic air monitoring.	Its coverage is limited and it is dependent on Wi-Fi connectivity.

Table 2.1: Comparison of Air Quality Monitoring Research Papers

Chapter 3

Methodology

The **purpose** of this chapter is to explain the methodological steps followed during the development of the IoT-based Air Quality Monitoring System. This project aims to measure temperature, humidity, and air quality using low-cost sensors connected to an ESP8266 microcontroller, with real-time data visualization on the Arduino IoT Cloud platform. This methodology outlines the systematic procedure followed from component integration to data acquisition and analysis.

Research design adopted for this work is an experimental design. This design is appropriate because the project requires building a working prototype, testing hardware components, collecting sensor readings, and evaluating system performance under different environmental conditions. The experimental approach ensures that each component—MQ135, DHT11, and ESP8266—is tested individually and together as a fully integrated system.

Quantitative research approach was used since the project primarily deals with numerical sensor data such as temperature, humidity, gas concentration values, and air quality levels. Quantitative measurement ensures objectivity and provides a clear basis for comparing results across different testing conditions.

The **sampling method** implemented in this project is purposive sampling. The system was tested in selected indoor environments where temperature and air quality variations could be observed clearly, such as rooms with moderate airflow and areas exposed to heat. Multiple tests were conducted at different intervals to ensure consistency.

Data collection was carried out using electronic sensors connected to the ESP8266 microcontroller. The DHT11 sensor captured temperature and humidity values, while the MQ135 sensor measured air quality by detecting gases such as ammonia, CO_2, smoke, and other impurities. The ESP8266 processed this data and transmitted it to the Arduino IoT Cloud through Wi-Fi, where the dashboard displayed real-time readings using gauges and text logs.

The **collected data** was analyzed using descriptive **analysis techniques**. Variations in temperature, humidity, and air quality were observed under normal and altered conditions. The IoT Cloud dashboard's numerical logs and live graphs assisted in identifying real-time changes, especially when heat exposure caused fluctuations in air quality values.

This project does not involve human subjects, personal data, or any sensitive information. All experiments were conducted in safe indoor environments, ensuring compliance with institutional and academic ethical guidelines.

The methodology is limited by the accuracy and sensitivity of low-cost sensors such as the MQ135 and DHT11, which may show minor variations over time or require periodic recalibration. Another **limitation** is the system's dependency on a stable Wi-Fi connection for transmitting data to the Arduino IoT Cloud.

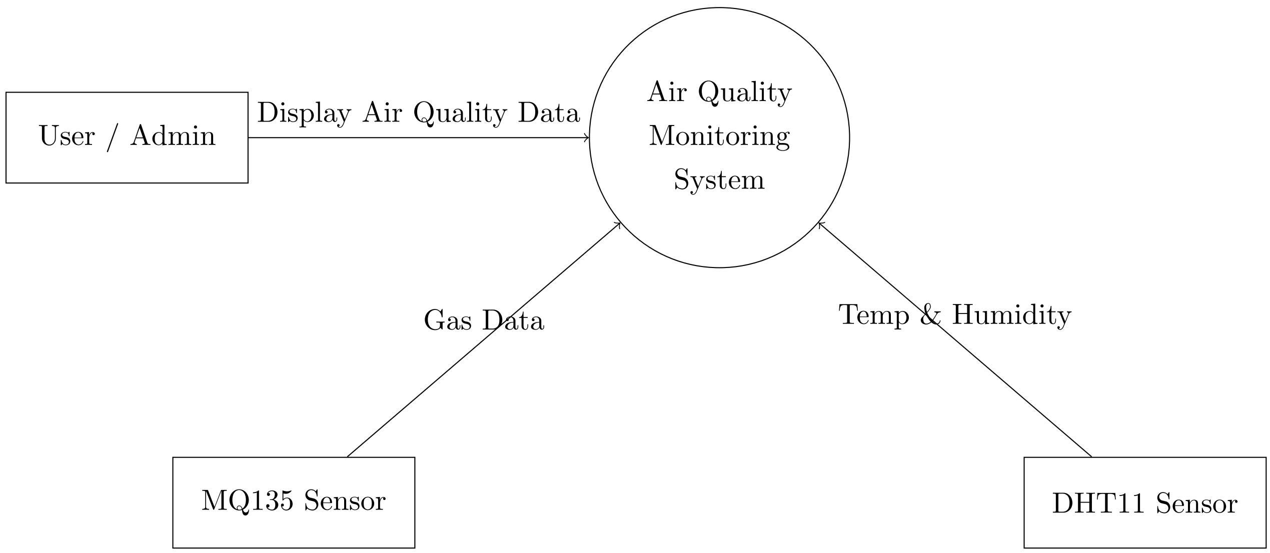

Figure 3.1: Data Flow Diagram of Air Quality Monitoring System (Level 0)

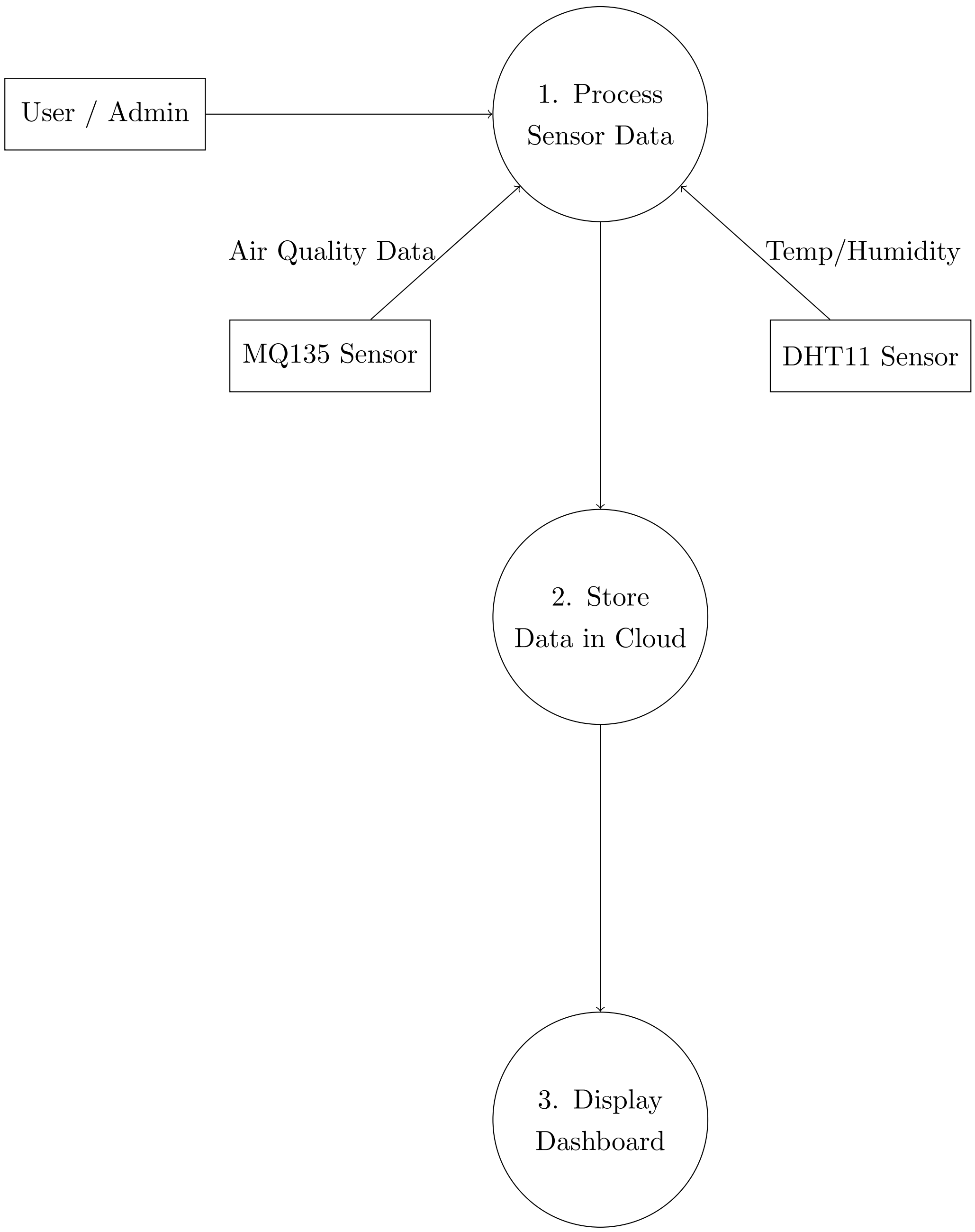

Figure 3.2: Data Flow Diagram of Air Quality Monitoring System (Level 2)

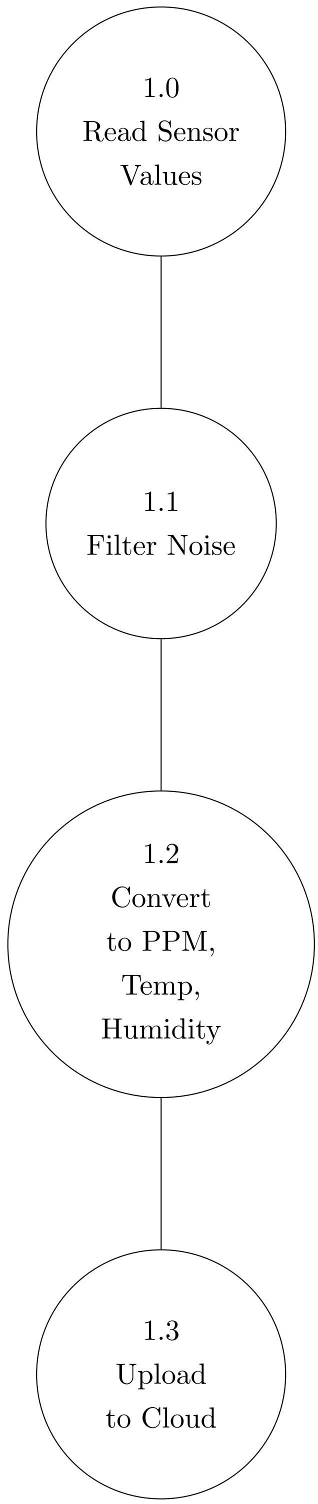

Figure 3.3: Data Flow Diagram of Air Quality Monitoring System (Level 3)

Chapter 4

Implementation

4.1 Introduction

This chapter presents the implementation of the IoT-based Air Quality Monitoring System including: a complete hardware setup, sensor interfacing, circuit designing, software development, and configuration of the Arduino IoT Cloud platform. Every step of the implementation was cautiously executed to have correct data acquisition and reliable real-time monitoring. All figures related to the implementation process are provided at the end of this chapter.

4.2 System Design Overview

The system design involves an ESP8266 NodeMCU microcontroller, which will collect environmental parameters through two sensors: the DHT11 sensor for temperature and humidity, and the MQ135 sensor for air quality measurement. The ESP8266 performs four major operations:

1. Read data from sensors
2. Processing raw sensor values
3. Connects to Wi-Fi via an onboard module.
4. Upload the processed values to Arduino IoT Cloud.

This cloud-based dashboard includes widgets for gauges, along with a live message log, so that end users can see data visualizations in real-time remotely.

4.3 Hardware Components

Hardware components used in this work are as follows:

- ESP8266 NodeMCU: This acts as the main controller and Wi–Fi module.
- **DHT11 Sensor:** It provides digital measurement of temperature and humidity.
- **MQ135 Gas Sensor:** It measures changes in surrounding air quality.
- Jumper wires, USB cable, and breadboard.
- External power supply (where necessary).

Each component has been selected based on cost-effectiveness, ease of interfacing, and accuracy for indoor air quality monitoring.

4.4 Hardware Interfacing

The ESP8266 provides the following digital and analog pins needed for sensor interfacing:

- The DHT11 sensor was connected to a digital pin (D4) of the NodeMCU.
- MQ135 was connected to the analog input pin A0.
- Both sensors were powered using the 3.3V and GND pins.

Proper wiring was maintained to avoid loose connections, and short jumper wires were used to minimize electrical noise. The MQ135 sensor has a warm-up time, so it was powered for several minutes before testing to obtain stable readings.

4.5 Circuit Description

The circuit design is simple and modular. Temperature and humidity are output as digital signals from the DHT11 sensor, while the MQ135 sensor outputs a voltage analogue to the concentration of such gases as NH_3, CO_2, and alcohol vapours.
The ESP8266 reads:

- Digital data from DHT11 via a single-wire protocol
- Analog voltage from MQ135 through its ADC pin

These values are processed and converted into meaningful real-time readings before being uploaded to the IoT Cloud.

4.6 Software Development

For writing software for the system, the Arduino IDE was used and it was extended by installation of the ESP8266 board support package. The following libraries were utilized:

- `DHT.h` to read from the DHT11 sensor
- `Arduino_IoT_Cloud.h` for cloud connectivity
- `WiFiConnectionManager.h` for Wi–Fi setup

The program does the following:

1. Initializes sensors and cloud variables
2. Connects the ESP8266 module to the configured Wi–Fi network
3. Continuously reads the values of temperature, humidity, and air quality
4. Sends data to the IoT Cloud dashboard periodically

The IoT Cloud contains three variables: **Temperature**, **Humidity** and **AirQuality**. These are mapped to the NodeMCU for automatic synchronisation.

4.7 Implementation Steps

The overall implementation process was carried out step by step:

1. Connected sensors to the ESP8266 microcontroller according to the circuit diagram.
2. Installed the necessary libraries and set up the ESP8266 board settings in Arduino IDE.
3. Wrote the program for reading sensor data and publishing it to Arduino IoT Cloud.
4. Created a Thing in the IoT Cloud and added cloud variables.
5. Linked the ESP8266 with the IoT Cloud device.
6. Connected NodeMCU to the Wi-Fi network to enable cloud connectivity.
7. Observed live data was provided on the dashboard through gauge displays and message logs.

4.8 Testing Procedure

The prototype was tested in three different environments:

- **Normal indoor room:** To evaluate baseline readings
- **Mild heat exposure:** To study system response to temperature changes.

The system was reliable in each scenario, and the readings were transmitted to the cloud instantly without any lag.

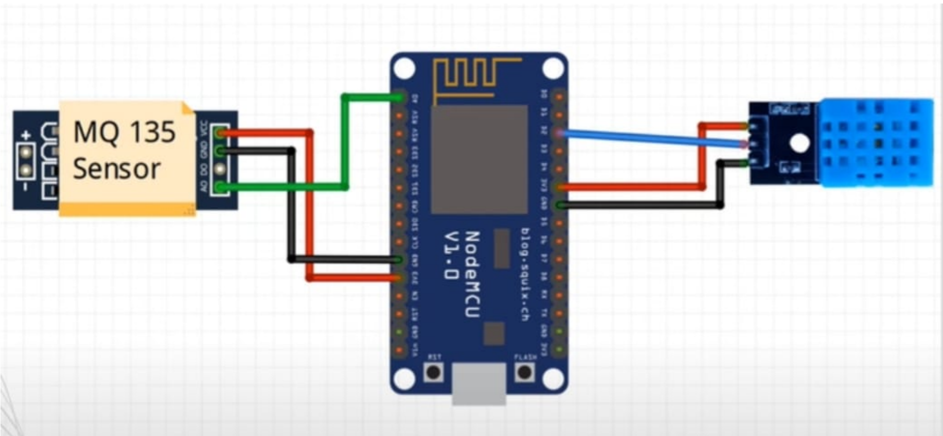

Figure 4.1: Hardware Connection Diagram

Circuit Connection

MQ 135 Sensor	ESP8266
A0	A0
GND	G
VCC	3V

DHT 11	ESP8266
Out	D2
GND	G
VCC	3V

Figure 4.2: Circuit Diagram

Chapter 5

Results and Analysis

5.1 Introduction

This chapter presents the results obtained from testing the IoT-based Air Quality Monitoring System. The readings were recorded in different environments to assess the behaviour and responsiveness of the sensors. Relevant dashboard screenshots and message logs are included for analysis.

5.2 Normal Environment Readings

The following ranges were observed under indoor room conditions:

- Temperature: 29°C to 35°C
- Humidity: 40
- Air Quality Value: 500–600

These are values for clean and stable room conditions. The dashboard readings were consistent and updated in real time.

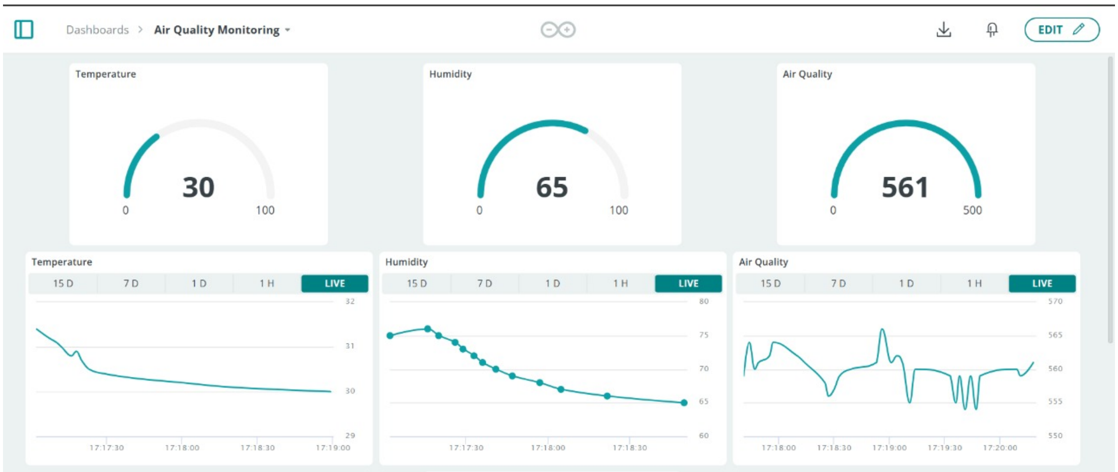

Figure 5.1: Normal Environmental Readings on IoT Dashboard

5.3 Heat Exposure Readings

The sensors were placed near a mild heat source to study their response. The following behavior was observed:

- Sharp rise in temperature
- Slight decrease in humidity
- Fall in air quality value due to changes in gas concentration.

These materials are not usually intended for general release. The system responded promptly and the changes were reflected on the dashboard instantly.

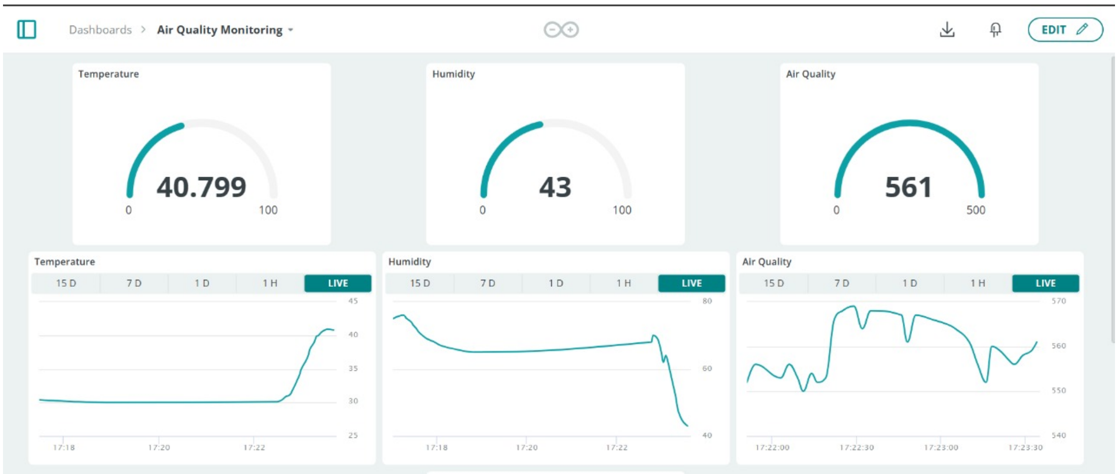

Figure 5.2: Readings After Heat Exposure

5.4 Message Log Analysis

These are, moreover, real-time message logs that are created within the Arduino IoT Cloud Dashboard, including sensor values. This would help track changes without always monitoring the continuous gauge widgets.

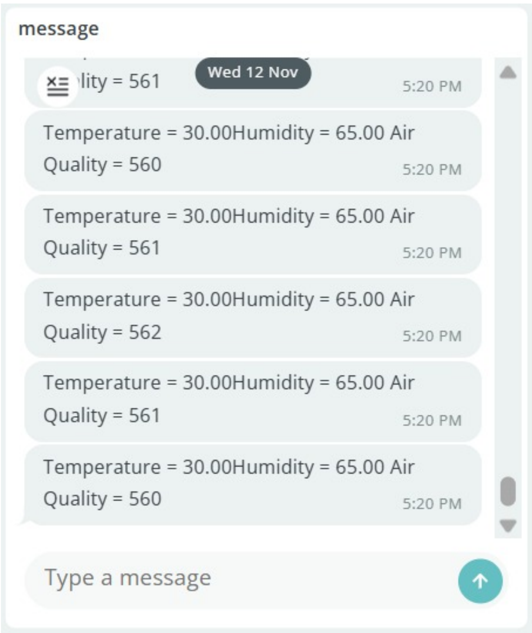

Figure 5.3: Real-Time Sensor Data Message – Sample 1

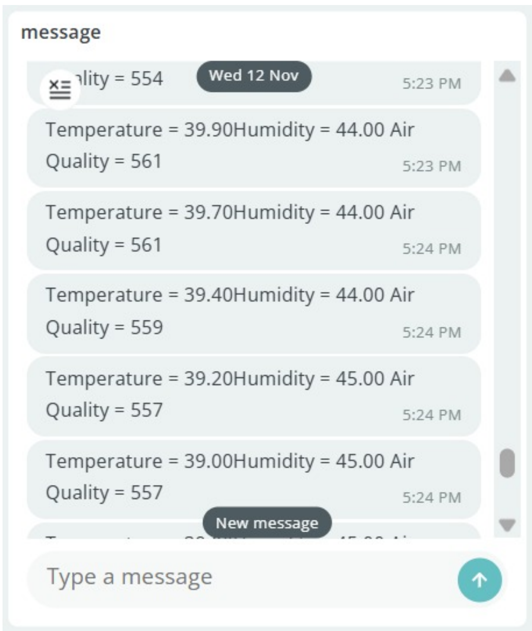

Figure 5.4: Real-Time Sensor Data Message – Sample 2

Chapter 6

Conclusion

The IoT-based Air Quality Monitoring System was implemented and tested, and the results showed that it is able to provide real-time data about environmental conditions reliably. Throughout the testing process, real-time readings of temperature and humidity, as well as air quality, were displayed.

- Under normal conditions, stable values were observed. The typical readings ranged from 29°C to 35°C for temperature, 40% to 70% for humidity, and 500 to 600 for air quality.
- When the MQ135 sensor was exposed to heat, temperature increased and the air quality values decreased instantly, showing rapid responsiveness of the sensor.
- The IoT Dashboard showed quick response and accurate data transmission, proving the capability for real-time monitoring by the system.
- The output of the real-time text message showed every sensor value clearly, thus supplying users with detailed numeric information.

Chapter 7

Future Scope

The project can be enhanced in several ways to improve functionality and expand practical applications. Potential future developments include:

- Integrating advanced gas sensors such as NO_2, CO, and SO_2 to enable more detailed and comprehensive air analysis.
- Introducing alert notifications through SMS or mobile push alerts to warn users immediately when air quality becomes unsafe.
- Incorporating Machine Learning (ML) techniques to support air quality prediction and trend forecasting.
- Adding voice assistant integration for enhanced user convenience and accessibility.
- Expanding the system into smart city environmental monitoring networks for large-scale deployment.

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